CSF2702: Object Oriented Design – Coursework

Due: 14:00am, 21st October 2026

You are part of a software development team that is creating a system for South Wales Hire and Haulage (SWHH), a vehicle hire and haulage company. Your team has already carried out a requirements analysis and now it is up to you to create a class diagram that models the classes and relationships between classes of the SWHH's hire system. The class diagram must have a fairly **low** level of abstraction (it should be detailed) and should contain enough information that it could be given to a programmer to implement. **You do NOT need to produce any code for this coursework.**

There are 25 marks available for this coursework and overall the coursework is worth 15% of the module.

An Important Note

Make sure that your model sticks to the problem as it is defined below. It can be tempting to think ahead and imagine extra or useful features in the system but please do not include these in your class diagram, you will not be graded for any additional or unnnessisary additions. Try to stick to the *business processes* involved.

Essential Functionaliy [20 marks]

Below are the requirements that you will need to work from. The systems needs to store information on customers, vehicles and rentals.

1. Each customer has various details associated with them such as names, contact information and drivers licence information.
2. A drivers license has a number, an expiry date, number of years held and categories of vehicles that can be driven.
3. The company is quite large and therefore hires out a variety of haulage vehicles. The three types of vehicle they hire are:
    - Articulated Lorries.
    - Luton Vans.
    - Standard Vans.
4. All vehicles have a fuel type, left or righthand drive and a gearbox type (automatic/manual), a maximum payload weight and a height. Standard Vans have the option of a tow bar attachment, Luton Vans have the option of a tail lift, and Articulated Lorries can optionally be hired out with a trailer.
5. The system should be able to add customers, remove customers, add vehicles, remove vehicles and produce a list of rentals that begin or end on a given date.
6. The system should be able to search for available vehicles by specifying the number of seats, the make or model or the available storage space (in m^3).
7. The system should be able to create a rental with the following information.
    - The vehicle that is being rented and its current mileage.
    - The date and time the rental begins.
    - How many days the rental is for.

- The location the car is being collected from and returned to.

8. A rental should also be able to be cancelled and a customer can only hire one vehicle.

You can assume that you have Date and Time data types and therefore do **not** need to create classes for them.

You should ensure that your design covers all of the above requirements and would be implementable ,in Java, if given to a programmer.

- Identifying sensible classes (and potentially subclasses) from the specification. [4 Marks]
- Well chosen attributes and useful operations. [4 Marks]
- Well chosen relationships between classes (e.g.,associations, aggregations, compositions, etc.). [4 Marks]
- Appropriate multiplicities that cover all reasonable requirements. [4 Marks]
- A design that would allow the system to operate correctly and cover all the requirements. [4 Marks]

Damage Reports [5 marks]

The company regularly deals with vehicles being returned with some damage and this can lead to disputes with the customers. They would like to enforce that every rental has at least two damage reports; one at the beginning of the rental and one at the end of a rental. They would like to store information on any exterior damage, interior damage and engine problems as textual descriptions. They would also like to record whether the damage needs to be fixed and a customer signature.

Customers should be able to check the damage reports associated with their rental and the company would like to be able to produce a list of damage reports that need fixing.

Marks will be awarded for a good, practical design that could be implemented.

Submission

You should submit a **single pdf document** via Canvas containing the following:

- The class diagram(s) embedded as **readable images** within the document.
- A description of each class and its responsibilities.

The document should have a logical structure with headings where appropriate. Please ensure that any screen shots are sufficiently readable since they will be printed on A4 paper for marking.

Submit your work via Canvas before **14:00am, 21st October 2026**.

By submitting coursework, electronically and/or hardcopy, you state that you fully understand and are complying with the University's policy on Academic Integrity and Academic Misconduct. The policy can be found at `https://myuni.swansea.ac.uk/academic-life/academic-misconduct`. *The consequences of committing academic misconduct can be extremely serious and may have a profound effect on your results and/or further progression. The penalties range from a written reprimand to cancellation of all of your marks and withdrawal from the University.*